Shivansh Nigam

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EDUCATION

Maulana Azad National Institute of Technology (MANIT) **Bhopal, MP, India**
Bachelor of Technology (B.Tech) in Electrical Engineering *Aug 2023 – May 2027*

- **Coursework:** Data Structures & Algorithms, AI/ML, DBMS, Operating Systems, Computer Networks, Linear Algebra.

TECHNICAL SKILLS

Languages & Core CS: Python, C++17, JavaScript, TypeScript, SQL, Data Structures & Algorithms, OOP, DBMS, OS
Generative AI & LLMs: LangChain, RAG Pipelines, AI Agents, Prompt Engineering, Llama 3, Gemini API, OpenAI API
Full-Stack & Backend: React.js, Next.js, Node.js, Express.js, FastAPI, RESTful APIs, WebSockets, Tailwind CSS
Databases & ML/Data: PostgreSQL, Redis, MongoDB, Scikit-learn, OpenCV, Pandas, NumPy, Explainable AI
Developer Tools & AI: Git, Docker, Postman, Linux/Bash, Cursor, Antigravity IDE, ChatGPT, Claude, GitHub Copilot

WORK EXPERIENCE

Yasa Ed Tech **Remote**
Software Engineering Intern *May 2026 – Jul 2026*

- Developed and integrated **10+ RESTful API endpoints** in Node.js, Express, and TypeScript with JWT role-based authentication (`Student` vs. `Admin`) and strict Zod schema validation for core learning modules.
- Optimized PostgreSQL query performance using composite indexes on `(user_id, course_id)` and implemented a Redis caching layer for course catalogs (15-min TTL), reducing redundant database reads by **40%**.
- Engineered an automated quiz-generation pipeline in Python using FastAPI and LLM prompt chaining to transform lesson markdown into structured JSON assessments, cutting manual content drafting time by **60%**.

PROJECTS

AgroMind – Context-Aware LLM Advisory & Explainable ML System [GitHub]
Tech Stack: Python, FastAPI, LangChain, Groq API (Llama 3), Scikit-learn, PostgreSQL, Docker

- Architected a context-aware agricultural advisory pipeline using **LangChain (LCEL)** and Groq Cloud LPUs (Llama 3 70B), streaming agronomic responses in under **800ms** by injecting 7 soil telemetry metrics (N, P, K, pH, moisture) and weather API telemetry.
- Trained an ensemble Random Forest model on 2,200+ soil-climate records across 22 crop classes, achieving **94.6% prediction accuracy** with **<20ms inference latency** and factor-by-factor Explainable AI (XAI) feature attribution.
- Developed 5 async FastAPI REST endpoints with dual PostgreSQL/SQLite ORM caching and containerized microservices via **Docker**, ensuring high availability and reproducible deployment.

SleekLink – Enterprise URL Redirection Engine & Analytics Dashboard [Live Demo] [GitHub]
Tech Stack: Node.js, Express, PostgreSQL, Redis, React, Vite

- Engineered a high-throughput URL redirection engine using Node.js and PostgreSQL, architecting a write-through Redis caching layer to resolve link lookups with **<1ms sub-millisecond latency** across **10K+ simulated requests/sec**.
- Architected an active/pause link state control with instantaneous Redis cache invalidation, eliminating stale-cache bypasses by immediately routing paused link traffic to a custom 403 Forbidden page.
- Developed a timezone-aware visitor analytics pipeline utilizing PostgreSQL `AT TIME ZONE` and a zero-dependency regex User-Agent parser to accurately extract device, OS, and referrer metrics.
- Built a responsive React 18 SPA dashboard featuring raw inline SVG charts, real-time client-side search filtering in **<5ms**, and virtual Blob downloaders for dynamic 300 DPI QR code generation.

VendorIQ – Vendor Performance Analytics & Intelligence Platform [GitHub]
Tech Stack: Python, Pandas, NumPy, SQL (SQLite CTEs), Streamlit, Git

- Architected an end-to-end data pipeline synthesizing **102K+ multi-table transactions** across 100 vendors into an indexed SQLite database, calculating complex aggregations in under **1 second**.
- Engineered multi-stage SQL Common Table Expressions (CTEs) to compute COGS, gross margins, and inventory ratios, validated by a **22-rule test suite** ensuring data integrity and mathematical invariants.
- Implemented a 4-quadrant rule-based vendor classification engine using dynamic median thresholds and deployed an interactive Streamlit dashboard featuring KPI metrics, search filters, and head-to-head comparisons.

Route Navigation Engine – Low-Latency C++ Pathfinding System [GitHub]
Tech Stack: C++17, Node.js, Express, Redis, HTML5 Canvas

- Engineered a high-performance pathfinding core in **C++17** implementing optimized A*, Dijkstra, BFS, and DFS on 10,000-node grid graphs, achieving **1.6 μs average calculation latency**.
- Reduced memory footprint by **40%** utilizing flat 1D vectors and bitsets; architected a hybrid backend combining Node.js Express sub-processes with Redis caching to resolve repeat routes in **<0.5ms**.
- Built an interactive HTML5 Canvas visualizer rendering smooth **60 FPS** path expansions, real-time obstacle placement, dynamic grid resizing, and instant start/target node re-computations.

ACHIEVEMENTS & LEADERSHIP

- **Technical Lead** at DataBiz (Data Science & Analytics Club), MANIT Bhopal.
- **Event Manager** at MAFFICK (Annual Cultural Fest), MANIT Bhopal.
- Solved **800+ Data Structures & Algorithms problems** across competitive platforms (Pupil at Codeforces).